30/08/2026

Code-E

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 **AIATS Prime for NEET-2027(XII Passed)_Part Test-01** **Time : 180 Min.**

Topics Covered:
Physics : Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion
Chemistry : Structure of Atom, Some Basic Concepts of Chemistry, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics
Botany : Cell: The Unit of Life, Cell Cycle and Cell Division, Biological Classification, The Living World, Morphology of Flowering Plants, Anatomy of Flowering Plants
Zoology : Structural Organisation in Animals - Animal Tissues, Biomolecules, Breathing and Exchange of Gases, Body Fluids and Circulation, Excretory Products and their Elimination, Locomotion & Movement

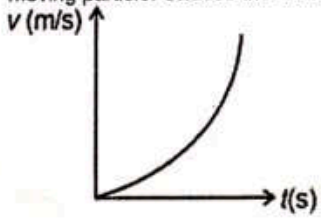
General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A car is moving with velocity of 20 m/s on a straight road. A scooterist wishes to overtake the car in 60 s. If the car is at a distance of 1.5 km ahead, then the constant velocity with which the scooterist has to chase the car is
   (1) 25 m/s
   (2) 20 m/s
   (3) 45 m/s
   (4) 50 m/s

2. Consider the following velocity(v) - time(t) graph of a moving particle. Choose the correct statement



   (1) Acceleration of the particle is constant
   (2) Acceleration of the particle is zero
   (3) Acceleration of the particle is increasing
   (4) Acceleration of the particle is decreasing

3. If $y = e^{3x}$, then the value of $\frac{dy}{dx}$ will be
   (1) e^{3x}
   (2) $3e^{3x}$
   (3) $\frac{e^{3x}}{3}$
   (4) $3e$

4. The moment of inertia of a rigid body is independent of
   (1) Mass of system
   (2) Distribution of mass in system
   (3) Position of axis of rotation
   (4) Angular speed of body

5. A shell is fired from a cannon at a speed of $240\ \text{m s}^{-1}$ in the direction $37°$ above the horizontal from the ground. The average velocity of shell in its entire journey is (Take $g = 10\ \text{m s}^{-2}$)
   (1) $144\ \text{m s}^{-1}$
   (2) $192\ \text{m s}^{-1}$
   (3) $162\ \text{m s}^{-1}$
   (4) $240\ \text{m s}^{-1}$

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6. **Assertion(A):** For a rigid body, separation between any two points changes under the application of forces on these two points.
Reason(R): For a rigid body rotating about a fixed axis, the angular velocity has a variable direction.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

7. A particle is moving in x-y plane. Position of the particle is given as $x = (4t^2 + 3)$ and $y = (3t^2 + 5)$ (where x and y are in metre and t in second). The magnitude of velocity of particles at $t = \frac{1}{2}$ second will be

(1) 7 m/s

(2) 5 m/s

(3) 6 m/s

(4) 3 m/s

8. Potential energy associated with a force is given as $U = (x^2 - 5x)$ J (where x is in meter). The magnitude of force at $x = 1$ m will be

(1) 1 N

(2) 7 N

(3) 4 N

(4) 3 N

9. The area of the acceleration-displacement curve of a body gives

(1) Impulse

(2) Change in momentum per unit mass

(3) Change in KE per unit mass

(4) Total change in energy

10. A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12. Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in m s^{-2} is:
($g = 10$ m/s^2)

(1) 1.8

(2) 1.2

(3) 1.5

(4) 2.1

11. The maximum speed with which a car can negotiate a curve of radius 10 m on an unbanked horizontal road of coefficient of friction 1 is ($g = 10$ m/s^2)

(1) 10 m/s

(2) 20 m/s

(3) 40 m/s

(4) 5 m/s

12. A person sitting on a rotating platform with arms stretched, suddenly folds his arm. Angular speed of platform increase. This is based on

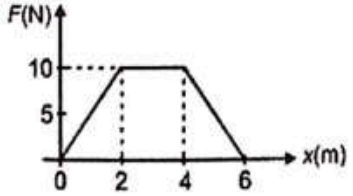
(1) Conservation of energy

(2) Conservation of momentum

(3) Conservation of angular momentum

(4) None of these

13. A particle of mass 1 kg is subjected to a force (F) which varies with position (x) according to F-x plot shown in figure. The change in kinetic energy from $x = 0$ to $x = 4$ m is



(1) 10 J

(2) 30 J

(3) 40 J

(4) 20 J

14. A body of mass 1 kg is moving on a horizontal circular path of radius 3 m with constant speed 6 m s^{-1}. The net force acting on the body is

(1) 10 N

(2) 6 N

(3) 12 N

(4) $\sqrt{244}$ N

15. Velocity (v) of a particle varies with time(t) as given below $v = At^2 + \frac{Bt}{C+t}$. The dimensions of $\frac{AB}{C}$ is

(1) LT^{-1}

(2) L^2T^{-5}

(3) L^2T^{-2}

(4) LT^{-5}

16. A vector is multiplied by a non-zero constant. The resulting vector has same magnitude but its direction is opposite to its initial direction. The value of the constant is

(1) –1

(2) 1

(3) –2

(4) 2

17. The coefficient of restitution 'e' for a perfectly inelastic collision is

(1) Zero

(2) 1

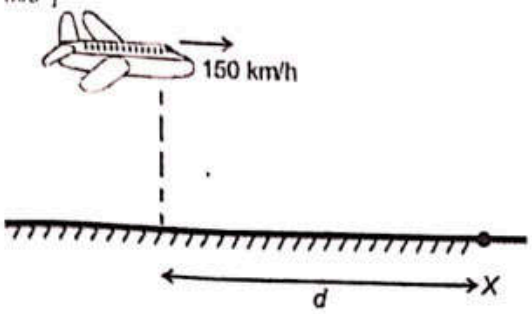
(3) –1

(4) ∞

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18. The airplane shown in the figure given below is in horizontal level height at an altitude of 500 m and having a speed of 150 km/h. At what approximate distance d should it release a heavy bomb to hit the target X ? [Take g = 10 m/s^2]



(1) 150 m

(2) 295 m

(3) 416 m

(4) 255 m

19. The position-time graph of two particles A and B are straight lines making angles of 37° and 30° respectively with time axis. If the velocity of A is v_A and that of B is v_B, then the value of $\frac{v_B}{v_A}$ is

(1) $\frac{4}{3}$

(2) $\frac{4}{\sqrt{3}}$

(3) 1

(4) $\frac{4}{3\sqrt{3}}$

20. A particle moves in a circle of radius 80 cm. Its linear speed increases as $v = 2t$ m s^{-1}, where t is in second. Net acceleration of the particle at t = 2 s will be

(1) $\sqrt{404}$ m/s^2

(2) $\sqrt{402}$ m/s^2

(3) $\sqrt{204}$ m/s^2

(4) $\sqrt{208}$ m/s^2

21. The area of plate is reported as 0.0006032 m^2, then the correct number of significant figure in the reported value will be

(1) 1

(2) 4

(3) 3

(4) 6

22. Two point masses of mass 1 kg and 2 kg are lying at point A(2, 0) m and B(–1, 0) m respectively. The co-ordinates of centre of mass is

(1) (0, 0) m

(2) $\left(\frac{1}{3}, 0\right)$ m

(3) $\left(\frac{1}{2}, 0\right)$ m

(4) $\left(-\frac{1}{2}, 0\right)$ m

23. A particle of mass $2m$ describes a circle of radius r. The centripetal acceleration of the particle is $\frac{1}{r^2}$. Angular Momentum of the particle about the centre is (all units are in SI)

(1) $2m\sqrt{r}$

(2) $m\sqrt{r}$

(3) $3m\sqrt{r}$

(4) $4m\sqrt{r}$

24. Which of the following is derived unit?

(1) Radian

(2) Steradian

(3) Newton

(4) Candela

25. **Statement I:** Dimensions of a physical quantity are the power to which the base quantities are raised to represent that quantity.
Statement II: The concept of dimensions can be used to find how various physical quantities depend on each other.

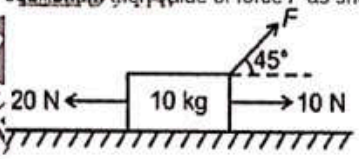
(1) Both statements I and II are incorrect

(2) Both statements I and II are correct

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

26. If the block placed on smooth surface is in horizontal equilibrium then value of force F as shown in the figure, is



(1) 10 N

(2) 20 N

(3) $10\sqrt{2}$ N

(4) $10\sqrt{3}$ N

27. A small ball of mass m moving with velocity v collides with an identical mass having velocity $\frac{v}{2}$ in same direction. If they stick together after collision, then velocity of centre of mass after collision is

(1) $\frac{v}{4}$

(2) $\frac{v}{2}$

(3) $\frac{3v}{4}$

(4) v

28. Work done by a conservative force in round trip of the system is always

(1) Zero

(2) Positive

(3) Negative

(4) Not finite

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29. Angle that the vector $\vec{A} = 2\hat{i} + 3\hat{j}$ makes with y axis is

(1) $\tan^{-1}\left(\frac{3}{2}\right)$

(2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\sin^{-1}\left(\frac{2}{3}\right)$

(4) $\cos^{-1}\left(\frac{2}{3}\right)$

30. The pitch of an error free screw gauge is 0.6 mm and there are 50 divisions on the circular scale. While measuring the diameter of a thin copper wire, the main scale reads 4 mm and the 40th division on the circular scale coincides with the reference line. The diameter of the wire is

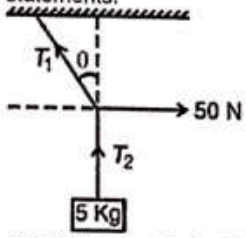
(1) 4.240 mm

(2) 4.480 mm

(3) 4.120 mm

(4) 4.400 mm

31. A mass of 5 kg is suspended by a rope of length 2 m from the ceiling. A horizontal force of 50 N is applied at mid point P of the rope, as shown. Read the following statements.



(a) The tension T_1 is 100 N
(b) The tension T_1 is $50\sqrt{2}$ N
(c) The angle θ is 45°
(d) The angle θ is 37°
The correct statement(s) are

(1) a and c only

(2) a and d only

(3) b and d only

(4) b and c only

32. A thin wire of mass $\frac{M}{2}$ and length L is bent to form a circular ring. The moment of inertia of ring about an axis passing through the centre and perpendicular to the plane is

(1) $\frac{ML^2}{4\pi^2}$

(2) $\frac{ML^2}{2\pi^2}$

(3) $\frac{ML^2}{8\pi^2}$

(4) $\frac{ML^2}{12}$

33. A projectile is thrown with initial velocity $\left(a\hat{i} + b\hat{j}\right)$ m/s. If range of projectile is equal to the maximum height reached by it then

(1) $b = a$

(2) $b = 2a$

(3) $b = 4a$

(4) $b = \frac{a}{2}$

34. The position of an object moving along straight line (x-axis) is given as $x = 2t + 3t^2$, where x in m and t in s. What is average velocity during first two second of motion?

(1) 16 m s^{-1}

(2) 8 m s^{-1}

(3) 4 m s^{-1}

(4) 14 m s^{-1}

35. A car moving with speed of 15 m s^{-1} can be stopped by applying brakes after travelling a distance 2 m. If the same car is moving with a speed of 7.5 m s^{-1}, the minimum stopping distance is (Assume same deceleration in both cases)

(1) 1 m

(2) 2 m

(3) 0.5 m

(4) 2.5 m

36. A bullet of mass 0.05 kg moving with speed of 80 ms^{-1} enters a wooden block and is stopped after a distance of 4 m. The average resistive force exerted by the block on the bullet is

(1) 300 N

(2) 20 N

(3) 400 N

(4) 40 N

37. A particle travels according to the equation $x = at^3$, $y = bt^3$. The equation of the trajectory is

(1) $y = \frac{ax^2}{b}$

(2) $y = \frac{bx^3}{a}$

(3) $y = \frac{bx^2}{a}$

(4) $y = \frac{bx}{a}$

38. A machine gun fires 10 bullets per second with speed 10 m/s. If mass of each bullet is 300 gm, then the force required to keep the gun stationary is

(1) 40 N

(2) 10 N

(3) 20 N

(4) 30 N

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39. A body of mass $\frac{1}{\sqrt{2}}$ kg moving along x-axis at 4 m/s, suddenly takes a right turn and starts moving along y-axis at the same speed. The magnitude of change in momentum of the body is

(1) 5 kg m/s

(2) $8\sqrt{2}$ kg m/s

(3) 4 kg m/s

(4) $2\sqrt{2}$ kg m/s

40. The velocity of an object is changing with time and the relation is given by the following equation $v = 3t^2 + 2t$. What is displacement of object in first 2 second?

(1) 18 m

(2) 12 m

(3) 10 m

(4) 22 m

41. The mass of box measured by a grocer's balance is 2.3 kg. Two gold pieces of masses 20.15 g and 20.17 g are added to the box. The total mass of the box now (to correct significant figure) is

(1) 2.300 kg

(2) 2.3 kg

(3) 2.34 kg

(4) 2.35 kg

42. The angular velocity of flywheel changes from 60 rpm to 120 rpm. Moment of inertia of flywheel is $\frac{5}{3}$ kg m^2. Change in angular momentum is (In SI unit)

(1) $\frac{5\pi}{3}$

(2) $\frac{2\pi}{3}$

(3) $\frac{10\pi}{3}$

(4) $\frac{8\pi}{3}$

43. In the following questions , a statement of assertion (A) is followed by a statement of reason (R).
A : If a body is accelerating then it must speed up.
R : Acceleration of body is equal to rate of change of speed.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

44. Match entries in column-I with entries in column-II and choose the correct option.

Column-I		Column-II	
A.	Limiting friction	P.	Has magnitude μ_k N
B.	Static friction	Q.	Maximum value of static friction
C.	Kinetic friction	R.	Is a self adjusting force

(1) A → P, B → Q, C → R

(2) A → R, B → P, C → Q

(3) A → Q, B → R, C → P

(4) A → Q, B → P, C → R

45. A body of mass m is attached to a vertical spring having spring constant k whose one end is fixed to ceiling. Body is allowed to fall under the gravity when spring is in its natural length. The maximum expansion in the spring will be

(1) $\frac{mg}{k}$

(2) $\frac{2mg}{k}$

(3) $\frac{mg}{2k}$

(4) Zero

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CHEMISTRY

46. Volume of $CO_2(g)$ obtained at STP by complete decomposition of 0.1 mol $NaHCO_3$ is

(1) 2 L

(2) 1.12 L

(3) 2.24 L

(4) 1.5 L

47. Mass of NaOH that must be added to water to prepare 500 mL of 2 M aqueous solution is

(1) 30 g

(2) 10 g

(3) 20 g

(4) 40 g

48. A compound contains 8% sulphur by mass, the minimum molecular mass of the compound is

(1) 800 u

(2) 1200 u

(3) 200 u

(4) 400 u

49. If the mass percentage of methanol in water is 16% then the molality of methanol in the solution is

(1) 7.25 m

(2) 3.21 m

(3) 8.27 m

(4) 5.95 m

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50. Among the following, maximum number of atoms are present in

(1) 14 g N_2

(2) 64 g SO_2

(3) 88 g CO_2

(4) 8 g H_2

51. 24 g of Mg will react completely with HCl to give

(1) 1 mole of H_2

(2) 2 moles of H_2

(3) 0.5 mole of H_2

(4) 2.5 moles of H_2

52. 100 ml of 0.2 M aqueous solution is mixed with 400 ml H_2O. The molarity of resultant solution is

(1) 0.1 M

(2) 0.05 M

(3) 0.025 M

(4) 0.04 M

53. 56 g of nitrogen and 20 g of hydrogen reacts in a closed vessel to form ammonia. Maximum amount of ammonia produced in this reaction is

(1) 17 g

(2) 34 g

(3) 68 g

(4) 51 g

54. Molarity of 100 dm^3 solution in which 0.2 mol NaOH is present is

(1) 0.2 M

(2) 0.02 M

(3) 0.002 M

(4) 0.0002 M

55. Consider the following statements:
Statement I: Mole fraction is temperature independent and unitless quantity.
Statement II: Number of moles of solute and molality of a solution are temperature independent.
In the light of above statements, choose the **correct** option.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

56. The ratio of radii of 2nd orbit of He^+ ion to 3rd orbit of Be^{3+} ion is

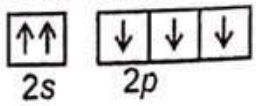
(1) 1 : 1

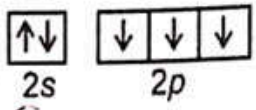
(2) 2 : 3

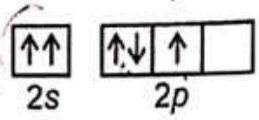
(3) 9 : 8

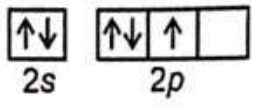
(4) 8 : 9

57. Which of the following orbital diagrams violates both Pauli's exclusion principle and Hund's rule?

(1) 

(2) 

(3) 

(4) 

58. What will be the work function of the metal (in eV), if energy of incident light is 6 eV and kinetic energy of photoelectron emitted is 3.2×10^{-19} J?

(1) 0.5 eV

(2) 4 eV

(3) 2 eV

(4) 1.5 eV

59. Number of electrons, protons and neutrons, respectively in phosphate anion are

(1) 50, 47 & 48

(2) 47, 48 & 50

(3) 42, 41 & 40

(4) 40, 41 & 42

60. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

61. Match column I with column II

Column I (Orbitals)	Column II (no. of angular nodes)
a. 1s	(i) 1
b. 2p	(ii) 3
c. 4d	(iii) 0
d. 4f	(iv) 2

The correct option is

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

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62. Ratio of the wavenumber of the first line of Lyman series of hydrogen atom to the first line of Balmer series of He^+ ion will be

(1) 5 : 2

(2) 9 : 5

(3) 27 : 20

(4) 7 : 5

63. Number of electrons present in a manganese atom having quantum numbers l = 1, m = 0 and s = $+\frac{1}{2}$ is

(1) 12

(2) 6

(3) 4

(4) 2

64. The de-Broglie wavelength of a particle with mass of 10 g and velocity 10 m/s is (h = 6.63×10^{-34} Js)

(1) 6.63×10^{-34} m

(2) 6.63×10^{-30} m

(3) 6.63×10^{-32} m

(4) 6.63×10^{-33} m

65. Number of electrons in chlorine atom having orbital angular momentum $\sqrt{2}\hbar$ is

(1) 6

(2) 5

(3) 11

(4) 8

66. Which among the following has lowest electronegativity?

(1) Cl

(2) Br

(3) F

(4) Na

67. Arrange Si, Be, Mg, Na and P in the increasing order of metallic character

(1) P < Si < Be < Mg < Na

(2) Na < Mg < Be < Si < P

(3) P < Si < Na < Mg < Be

(4) Si < P < Mg < Na < Be

68. An atom has electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$. You will place it in which group of modern periodic table?

(1) 4

(2) 5

(3) 6

(4) 7

69. The general electronic configuration of *p*-block elements is

(1) $ns^{1-2}np^{0-6}$

(2) $ns^{0}np^{1-6}$

(3) $ns^{2}np^{0-6}$

(4) $ns^{2}np^{1-6}$

70. Consider the following pair of elements
(a) N and S
(b) Mg and B
(c) Li and Mg
(d) Be and Al
The pairs of elements which are diagonally related are

(1) (a) and (b) only

(2) (c) and (d) only

(3) (b), (c) and (d) only

(4) (a), (c) and (d) only

71. Eka – aluminium and Eka - silicon in Mendeleev's periodic table respectively are

(1) Ga, Ta

(2) Ga, Sn

(3) Ge, Ga

(4) Ga, Ge

72. The correct order of first ionization enthalpy for the given elements is

(1) Be < B < C < N < O < F

(2) B < Be < C < O < N < F

(3) Be < B < C < O < N < F

(4) B < Be < C < N < O < F

73. Consider the following statements.
(i) PCl_5 has trigonal bipyramidal shape.
(ii) The equatorial bonds in PCl_5 are longer than those of the axial bonds.
(iii) 90° bond angle (Cl – P – Cl) in PCl_5 is absent.
Choose the incorrect statement.

(1) (i) only

(2) (ii) and (iii) only

(3) (ii) only

(4) (i), (ii) and (iii)

74. Match List I with List II :

	List I		List II
A.	C_2H_4	I.	3 σ bonds, 2 π bonds
B.	C_2H_2	II.	3 σ bonds, one lone pair
C.	CH_4	III.	4 σ bonds
D.	NH_3	IV.	5 σ bonds, 1 π bond

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-I, C-III, D-II

(3) A-I, B-II, C-IV, D-III

(4) A-II, B-III, C-I, D-IV

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75. The ratio of σ-bond and π-bond in $C_2(CN)_4$ is

(1) 2 : 1

(2) 1 : 1

(3) 1 : 2

(4) 3 : 4

76. The formal charge on the central oxygen atom in O_3 molecule is

(1) +1

(2) –1

(3) 0

(4) +2

77. Which of the following is an exception to the octet rule?

(1) PCl_3

(2) SCl_2

(3) SF_6

(4) CO_2

78. Species in which all bond lengths are not equal is

(1) O_3

(2) SF_6

(3) SF_4

(4) CCl_4

79. Select the polar molecule among the following

(1) NF_3

(2) CF_4

(3) CO_2

(4) SF_6

80. Maximum dipole moment among the following is of

(1) NF_3

(2) NH_3

(3) BF_3

(4) CCl_4

81. Match List I with List II.

List I Processes	List II Conditions
(a) Adiabatic	(i) $\Delta V = 0$
(b) Isochoric	(ii) $q = 0$
(c) Isothermal	(iii) $\Delta P = 0$
(d) Isobaric	(iv) $\Delta T = 0$

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(i), d(ii)

82. 2 moles of a gas is expanded isothermally and reversibly from 10 lit. to 50 lit at 300 K temperature. Work done during the process is (Given $\log_{10}5 = 0.7$)

(1) – 8.04 kJ

(2) + 12.5 kJ

(3) – 4.01 kJ

(4) + 6.2 kJ

83. If enthalpy of atomization of methane is 1665 kJ, what is the bond energy of C–H?

(1) 832.5 kJ

(2) 416.25 kJ

(3) 3330 kJ

(4) 6660 kJ

84. Match column-I with column-II

Column-I (Reaction)	Column-II (Relation between ΔH and ΔU)
a. $C_2H_5OH\ (l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O\ (l)$	(i) $\Delta H = \Delta U - 3RT$
b. $Fe_2O_3(s) + 3H_2(g) \rightarrow 2Fe(s) + 3H_2O\ (l)$	(ii) $\Delta H = \Delta U + \frac{RT}{2}$
c. $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$	(iii) $\Delta H = \Delta U - RT$
d. $CO_2(g) \rightarrow CO(g) + \frac{1}{2}O_2(g)$	(iv) $\Delta H = \Delta U + 2RT$

Choose the correct answer from the options given below.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

85. For the gas phase reaction

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

Which of the following conditions are correct?

(1) $\Delta H < 0$ and $\Delta S < 0$

(2) $\Delta H > 0$ and $\Delta S < 0$

(3) $\Delta H > 0$ and $\Delta S > 0$

(4) $\Delta H = 0$ and $\Delta S < 0$

86. Consider the following statements

(a) Heat capacity, pressure and temperature are intensive properties

(b) Standard molar enthalpy change for vapourisation of NH_3 is higher than N_2.

(c) Standard molar enthalpies of formation of $Cl_2(g)$, $H_2(g)$ and $Br_2(g)$ are zero at 298 K.

The correct statement(s) is/are

(1) (b) only

(2) (a) and (b) only

(3) (a), (b) and (c)

(4) (b) and (c) only

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87. Reaction for which ΔH is equal to ΔU, is

(1) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

(2) $C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$

(3) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

(4) $C(s) + O_2(g) \rightarrow CO_2(g)$

88. At 298 K, which of the following reactions represent(s) enthalpy of formation?

(1) $CO(g) + 2\,H_2(g) \rightarrow CH_3OH(l)$

(2) $2\,C(graphite) + 3\,H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$

(3) $C_2H_2(g) + \frac{5}{2}O_2(g) \rightarrow 2\,CO(g) + H_2O(l)$

(4) $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$

89. The correct thermodynamic conditions for a non-spontaneous reaction at all temperature is

(1) $\Delta H < 0$ and $\Delta S > 0$

(2) $\Delta H < 0$ and $\Delta S = 0$

(3) $\Delta H > 0$ and $\Delta S < 0$

(4) $\Delta H > 0$ and $\Delta S > 0$

90. The entropy of any pure crystalline substance approaches zero as the temperature approaches

(1) 0°C

(2) 0 K

(3) 100°C

(4) 273°C

BOTANY

91. Mesosome helps in all, **except**

(1) Cell wall formation

(2) DNA replication

(3) Formation of nucleolus

(4) Respiration

92. The plastid which is colourless and stores proteins is

(1) Amyloplast

(2) Aleuroplast

(3) Elaioplast

(4) Chloroplast

93. Read the following statements and choose the **correct** option accordingly.
Statement A : Several ribosomes may attach to a single mRNA and form a chain called polyribosomes.
Statement B : The number of microtubules present in the cross-section of an axoneme of flagella is more than that in centriole.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

94. Algal cell wall is composed of
(a) Galactans
(b) Calcium carbonate
(c) Mannans
(d) Cellulose
The **correct** one(s) is/are

(1) All (a), (b), (c) and (d)

(2) Only (d)

(3) Only (a), (c) and (d)

(4) Only (a) and (c)

95. Read the following statements of **Assertion (A)** and **Reason (R)**.
Assertion (A): Cell is the fundamental, structural and functional unit of all living organisms.
Reason (R): Anything less than a complete structure of a cell does not ensure an independent living.
In the light of above statements select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

96. State the following statements as True **(T)** or False **(F)** regarding the cell membrane and select the **correct** option.
(a) In human beings, the membrane of the erythrocyte has approximately 40 percent lipids and 52 percent proteins.
(b) Integral and peripheral proteins lie on the surface of the membrane.
(c) Quasi-fluid nature of lipids enables lateral movement of proteins within the overall bilayer.
(d) The fluid nature of the membrane helps in functions like cell growth, cell division and formation of intercellular junctions.

(1) (a)-T, (b)-T, (c)-F, (d)-T

(2) (a)-T, (b)-F, (c)-T, (d)-T

(3) (a)-T, (b)-F, (c)-F, (d)-F

(4) (a)-F, (b)-T, (c)-T, (d)-T

97. Kinetochores are

(1) Terminal ends of chromosomes

(2) Present around primary constriction and help in crossing over

(3) Present near nuclear envelope and take part in spindle formation

(4) Present around centromere and provide a site for attachment to the spindle fibers

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98. Identify the statement which is **NOT** true regarding the cellular structures that are extensive and continuous with the outer membrane of the nucleus.

(1) They bear ribosomes on their surface

(2) These are frequently observed in the cells actively involved in protein synthesis and secretion

(3) They give rise to structures responsible for detoxification of drugs

(4) They form plasma membrane during cell division

99. Golgi apparatus is the important site of formation of

(1) Glycerol and amino acids

(2) Glycolipids and glycoproteins

(3) Glucose and Galactose

(4) Glycerides and glycine

100. Spindle fibers attach to kinetochores of chromosomes during

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

101. The content of DNA doubles during the

(1) G_1 phase

(2) S-phase

(3) G_0 phase

(4) G_2 phase

102. The end stage of karyokinesis is

(1) Telophase

(2) Prophase

(3) Metaphase

(4) Anaphase

103. The most dramatic period of the cell cycle is

(1) G_1 phase

(2) M phase

(3) G_2 phase

(4) S phase

104. Read the following statements and select the **correct** option.
Statement A: Karyokinesis is not followed by cytokinesis in the cells of coconut water.
Statement B: Cells in quiescent stage remain metabolically active but do not proliferate unless called on to do so.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

105. Consider the following statements related to cell cycle, and select the **correct** option stating them as True [T] or False [F].
(A) Nucleolus, ER, chromosomes, Golgi complex, nuclear envelope are reformed during telophase stage.
(B) The number of chromosomes doubles during S-phase.
(C) Events of cell cycle are independent of genetic control.
(D) In the S and G_2 phases the new DNA molecules formed are not distinct but intertwined
(E) Acidic conditions are maintained inside the lysosome by pumping of H^+ ions into them.

(1) A-(T), B-(T), C-(F), D-(F), E-(F)

(2) A-(F), B-(F), C-(F), D-(T), E-(T)

(3) A-(T), B-(F), C-(T), D-(F), E-(T)

(4) A-(F), B-(F), C-(T), D-(T), E-(T)

106. Select the **incorrect** statement regarding interkinesis.

(1) It is short-lived

(2) Chromosomes elongate but do not form chromatin fibres

(3) It occurs in diploids cells therefore no DNA replication is required

(4) Occurs between telophase I and prophase II

107. Mitotic apparatus is composed of
(a) Asters
(b) Microtubules
(c) Spindle fibres
(d) Centromere
(e) Kinetochore
Choose the **correct** one (s).

(1) a, b and e only

(2) a, b and d only

(3) a, b and c only

(4) a, b, c and d only

108. What is/are **true** about diakinesis of prophase-I?
a. It represents transition to metaphase-I
b It is final stage of prophase-I
c. Chiasmata formation occurs
d. Nucleolus reappear

(1) Only b

(2) Only a and b

(3) Only b and c

(4) All a, b, c and d

109. Identify the **correctly** matched pair.

(1)	*Triticum*	–	Placed in the division Monocotyledonae
(2)	Alsatians	–	These are placed in the family felidae with cats
(3)	Gibbon	–	Placed in the class Carnivora
(4)	Insects	–	Placed in phylum Arthropoda

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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110. Systematics is different from taxonomy as it takes into account

(1) Characterisation

(2) Identification

(3) Nomenclature

(4) Evolutionary relationships

111. Book 'Systema Naturae' for animals was written by

(1) Ernst Mayr

(2) Carolus Linnaeus

(3) Ernst Haeckel

(4) Robert Hooke

112. Mixotrophic mode of nutrition can be seen in

(1) Dinoflagellates

(2) Euglenoids

(3) Protozoans

(4) Chrysophytes

113. *Trypanosoma*, *Chlamydomonas* and *Gonyaulax* show a common feature that is

(1) Presence of cellulosic wall and predominance of inclusion bodies

(2) Presence of photosynthetic pigment

(3) Presence of flagella and are placed in same kingdom according to R.H. Whittaker

(4) Similar body organisation and always found in freshwater

114. The parasitic fungi on mustard is

(1) *Mucor*

(2) *Rhizopus*

(3) *Albugo*

(4) *Neurospora*

115. Select the **odd** one w.r.t. *Nostoc*.

(1) Can fix atmospheric nitrogen

(2) Perform anoxygenic photosynthesis

(3) Covered by mucilagenous sheath

(4) Have a rigid cell wall

116. Read the following statements.
(a) Cellulosic cell wall is present in the members of the kingdom Plantae.
(b) Viroids were discovered by Dmitri Ivanowsky.
(c) Commonly known forms of the class basidiomycetes are mushrooms, morels and truffles.
(d) Chief producers in oceans are responsible for the formation of diatomaceous earth.
The **correct** one(s) is/are:

(1) (a), (b) and (d)

(2) (a) and (d) only

(3) (b) and (c) only

(4) (b) only

117. The organism that causes sleeping sickness differs from the malarial parasite as the former

(1) Belongs to the kingdom Protista

(2) Possesses flagella as locomotory structures

(3) Is heterotrophic in the mode of nutrition

(4) Has an infectious spore-like stage in its life cycle.

118. Identify the **correct** statement w.r.t. the kingdom Monera.

(1) Its sole members are grouped under a total of three categories on the basis of their shapes.

(2) All the members are characterised by the presence of a rigid cell wall.

(3) Vast majority of its members depend on other organisms or dead organic matter for their nourishment.

(4) Its members play a very little role in the recycling of nutrients.

119. The causative agent of the potato spindle tuber disease is

(1) A nucleoprotein

(2) Infectious RNA particle with low molecular weight

(3) A symbiotic association between algae and fungi

(4) An abnormally folded protein

120. Match List I with List II.

	List-I		List-II
A.	*Rhizopus*	I.	Mushroom
B.	*Ustilago*	II.	Smut fungus
C.	*Puccinia*	III.	Bread mould
D.	*Agaricus*	IV.	Rust fungus

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-IV, B-III, C-II, D-I

121. In china rose
(a) Petals show twisted aestivation
(b) Ovary is superior
(c) Diadelphous stamens are found
(d) Placenta is present at the periphery of ovary

(1) Only (a) and (d) are true

(2) Only (a) and (b) are true

(3) (b), (c) and (d) are true

(4) (a), (b) and (d) are true

122. In which of the following plants, margin of sepals or petals overlaps each other but not in a particular direction?

(1) Chilli

(2) Cotton

(3) *Cassia*

(4) *Calotropis*

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123. Observe the following diagrams:

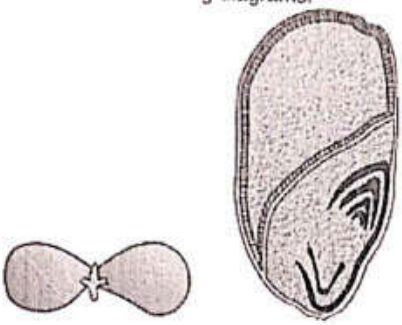


Figure (A) Figure (B)

Select the option that correctly describes the difference between these two types of seed.

	Figure A	Figure B
(1)	This type of seed has a small pore called micropyle	This type of seed lacks the small pore called micropyle
(2)	Plants having this type of seed generally have tap roots and reticulate venation	Plants having this type of seed generally have fibrous roots and parallel venation
(3)	In this type of seed, the embryonal axis is covered with the undifferentiated sheaths	In this type of seed, embryonal axis is not covered with the undifferentiated sheaths
(4)	Plant having this type of seed requires 2 more ATP per CO_2 fixed than a C_3 plant.	Plant having this type of seed can only show CAM pathway

(1) (1)

(2) (2)

(3) (3)

(4) (4)

124. Identify the **correct** floral formula of a family to which belladonna belongs.

(1) $Br \oplus \female \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$

(2) $\oplus \female K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$

(3) $\% \female K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$

(4) $\oplus \female K_{2+2} C_{x} A_{2+4} \underline{G}_{(2)}$

125. Ground tissue system consists of

a. Parenchyma
b. Collenchyma
c. Sclerenchyma

(1) Only a

(2) b and c only

(3) b only

(4) All a, b and c

126. Examine the following structures and identify the haploid, diploid and triploid structures, respectively.

(a) Endosperm of castor
(b) Seed coat of pea
(c) Embryo sac of *Hibiscus*
(d) Aleurone layer of wheat
(e) Scutellum of maize
(f) Radicle of *Petunia*

	Haploid	Diploid	Triploid
(1)	(b), (e)	(c), (f)	(a), (d)
(2)	(c), (e)	(d)	(a), (b), (f)
(3)	(c)	(b), (e), (f)	(a), (d)
(4)	(a), (b)	(d), (e)	(c), (f)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

127. The leaflets are attached at a common point, i.e., at the tip of the petiole in

(1) Neem

(2) Guava

(3) *Calotropis*

(4) Silk cotton

128. In dicot stems conjoint vascular bundles are open, because

(1) There is presence of vascular cambium

(2) Arrangement of xylem and phloem at different radii

(3) Presence of phloem at outer side of xylem

(4) Presence of xylem at periphery

129. Initiation of lateral roots and vascular cambium during secondary growth takes place in cells of

(1) Pericycle

(2) Epiblema

(3) Cortex

(4) Endodermis

130. Root hair and trichomes both are

(1) Always unicellular

(2) Branched

(3) Epidermal outgrowths

(4) Responsible for preventing water loss

131. Calyx and corolla are not distinct in flowers of

(1) Lily

(2) Mustard

(3) Pea

(4) Brinjal

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132. Read the given statements of assertion (A) and reason (R) and choose the **correct** option.
Assertion (A): Stems of maize and wheat do not possess the ability to form secondary xylem and secondary phloem tissues.
Reason (R): In stems of maize and wheat, the xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) Both (A) and (R) are false.

(4) (A) is true but (R) is false.

133. Select the option for **incorrect** set of statements w.r.t. subsidiary cells.
(i) These are specialised epidermal cells.
(ii) These are also called bulliform cells.
(iii) They are found between the two guard cells of a stomata.
(iv) They do not carry out photosynthesis.

(1) (i) and (iii)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) (ii) and (iv)

134. Semi-lunar patches of sclerenchyma is present

(1) On the inner side of endodermis and above the protoxylem in dicot root.

(2) On the inner side of the endodermis and above the phloem in dicot stem.

(3) On the inner tangential wall of the inner cortex of dicot stem.

(4) In between the xylem and phloem present on the same radii.

135. Read the given statements and state them as True(**T**) or False(**F**).
(A) In leaf, stem and root of both sunflower and maize, the epidermis is often covered with a waxy thick layer and possesses unicellular or multicellular hairs.
(B) The outermost layer of stele in roots of pea and mustard shows depositions of suberin in the form of strips.
(C) Variation in the size of vascular bundles, presence of water containing cavities within vascular bundles are characteristics of monocot stem.
(D) The outermost layer in cortex of dicot stem is composed of living mechanical tissue.
Choose the **correct** option.

(1) (A)-F, (B)-T, (C)-F, (D)-T

(2) (A)-T, (B)-F, (C)-T, (D)-F

(3) (A)-F, (B)-F, (C)-T, (D)-T

(4) (A)-T, (B)-T, (C)-F, (D)-T

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136. The function of cilia present in inner lining of bronchioles is to facilitate

(1) Diffusion of gases

(2) Movement of mucus

(3) Absorption of mucus

(4) Secretion of mucus

137. Fusiform muscle fibres, that do not show striations and are involuntary will be found primarily in

(1) Limbs

(2) Wall of stomach

(3) Wall of heart

(4) Diaphragm

138. For ash analysis, living tissue is first dried and then burnt. The ash obtained subsequently contains

(1) Both organic and inorganic compounds

(2) Similar compounds which were present in acid insoluble pool while conducting compound analysis of a living tissue

(3) Only inorganic compounds/elements

(4) Only organic compounds

139. Brush bordered cuboidal epithelium is present in the lining of

(1) Proximal convoluted tubules

(2) Small intestine

(3) Stomach

(4) Smaller ducts of pancreas

140. Select the odd one w.r.t. the presence of collagen fibres.

(1) Tendons

(2) Outer ear joints

(3) Beneath the skin

(4) Lymph present in lymph vessels

141. The body of a simple organism like *Hydra* is made of

(1) A single cell

(2) Different types of cells

(3) Only single type of multiple cells

(4) Only two types of cells

142. Which of the following is an acidic amino acid?

(1) Glycine

(2) Serine

(3) Cysteine

(4) Glutamic acid

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143. Comprehend the given statements w.r.t. starch and choose the incorrect one.

(1) It forms helical secondary structure.

(2) It can hold I_2 molecules in the helical portion.

(3) The starch-I_2 complex is red in colour.

(4) It is present as a store house of energy in plant tissues.

144. The steps involved in a catalytic cycle of an enzyme are given below:
(a) Enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate
(b) Alteration in the shape of an enzyme to fit the substrate more tightly
(c) Binding of substrate to the active site of the enzyme
(d) Breakage in the chemical bonds of the substrate and formation of new enzyme-product complex
Arrange the above in the correct chronological order. Select the **correct** option.

(1) c → d → b → a

(2) c → b → d → a

(3) a → b → c → d

(4) c → a → b → d

145. Cytidine differs from uridylic acid as

(1) Former is the monomeric unit of DNA while latter is present in RNA

(2) Phosphoric acid is a component of former only

(3) Former possesses three N-atoms in its nitrogenous base

(4) Former possesses two heterocyclic rings in its structure

146. In a molecule of glycogen, left end has

(1) Free aldehydic group representing reducing end

(2) Free carboxyl group representing C-terminal amino acid

(3) No free aldehydic group, representing non-reducing end

(4) Free ketonic group representing N-terminal amino acid

147. Match Column-I with Column-II and choose the **correct** option.

	Column I		Column II
a.	Beta-pleated sheet	(i)	Simplest amino acid
b.	Glycosidic bond	(ii)	Present in tertiary and quaternary structure of proteins
c.	Disulphide bond	(iii)	Exist between two simple sugars in oligosaccharide
d.	Glycine	(iv)	Secondary structure of protein

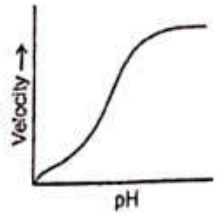
(1) a(i), b(ii), c(iii), d(iv)

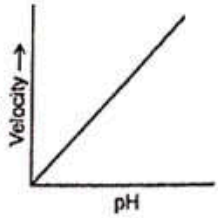
(2) a(iv), b(iii), c(ii), d(i)

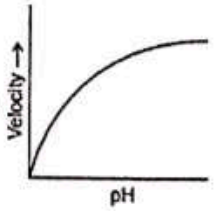
(3) a(iv), b(ii), c(iii), d(i)

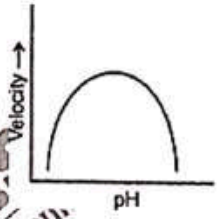
(4) a(iv), b(i), c(iii), d(ii)

148. Which of the following graph represents **correctly** the effect of pH on the velocity of a typical enzymatic reaction?

(1)


(2)


(3)


(4)


149. All of the given elements form higher per cent weight of Earth's crust in comparison to human body, **except**

(1) Sodium

(2) Carbon

(3) Magnesium

(4) Calcium

150. **Assertion (A):** Fats remain solid at room temperature while oils remain liquid at room temperature.
Reason (R): Fats have a lower melting point, while oils have a higher melting point.

(1) Both (A) and (R) are true but (R) is not the corect explanation of (A)

(2) Both (A) and (R) are false

(3) (A) is true (R) is false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

151. How many layers majorly contribute in the formation of a respiratory membrane in lungs of humans?

(1) One

(2) Three

(3) Two

(4) Six

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152. Lungs are present in thoracic chamber which is ventrally, dorsally and laterally formed respectively by

(1) Vertebral column, ribs, sternum

(2) Ribs, sternum, diaphragm

(3) Sternum, vertebral column, ribs

(4) Diaphragm, sternum, ribs

153. The enzyme which catalyses the following reaction

$CO_2 + H_2O \overset{E}{\rightleftharpoons} H_2CO_3 \overset{E}{\rightleftharpoons} H^+ + HCO_3^-$

is present

(1) Only in RBCs

(2) Only in blood plasma

(3) In cytoplasm of all body cells

(4) Mainly in RBCs and minute quantities in plasma

154. The factor that shifts the oxygen-hemoglobin dissociation curve to the left side is

(1) High partial pressure of oxygen

(2) High partial pressure of carbon dioxide

(3) High body temperature

(4) High H^+ concentration

155. Which one of the following is **correct** option about the transport of CO_2 by deoxygenated blood, under normal condition?

	Dissolved in Plasma form	Bicarbonate ions form	Carbamino-haemoglobin form
A.	23%	70%	7%
B.	7%	70%	23%
C.	70%	23%	7%
D.	3%	23%	70%

(1) A

(2) B

(3) C

(4) D

156. If the cardiac output of a person is 4900 mL and blood volume in the left ventricle at the end of diastole is 140 mL while at the end of ventricular systole is 70 mL, then what is the heart rate of this person?

(1) 75 beats per minute

(2) 70 beats per minute

(3) 35 beats per minute

(4) 140 beats per minute

157. The volume of air that remains in the lungs after a normal expiration is called

(1) Expiratory Reserve Volume

(2) Expiratory Capacity

(3) Functional Residual Capacity

(4) Residual Volume

158. The structure starting with the external nostrils up to the terminal bronchioles constitute the

(1) Respiratory part

(2) Exchange part

(3) Conducting part

(4) Reducing part

159. Select the **mismatch** w.r.t. formed elements.

(1) Thrombocytes – Clotting of blood

(2) Eosinophils – Allergic reactions of the body

(3) Lymphocytes – Participate only in phagocytosis

(4) Basophils – Inflammatory reactions of the body

160. Neutrophils and monocytes differ from each other in all of the following aspects, **except**

(1) The shape of their nuclei in mature form

(2) Their contribution to total WBC content of blood

(3) Their mode of action against foreign organisms

(4) Their cytoplasmic contents

161. Select the **correct** option w.r.t. the features of human heart.

(1) Endodermal in origin

(2) Functions can be moderated by endocrine system

(3) Slightly tilted towards the right side of the body

(4) Protected by a single layer of membranous pericardium

162. Select the **incorrect** statement among the following

(1) During ventricular systole, tricuspid and bicuspid valves get closed

(2) During joint diastole, tricuspid and bicuspid valves remain open

(3) During ventricular diastole, semilunar valves open

(4) The SAN generates an action potential which stimulates both atria causing atrial systole

163. Which of the following is the **correct** sequence for conducting pathway of impulses in human heart?

(1) SA node → Purkinje fibres → Bundle of His → AV node → Ventricular muscles

(2) AV node → Bundle of His → SA node → Purkinje fibres → Ventricular muscles

(3) SA node → AV node → Bundle of His → Purkinje fibres → Ventricular muscles

(4) SA node → AV node → Purkinje fibres → Bundle of His → Ventricular muscles

164. Choose the **correct** option to complete the analogy.
Pulmonary vein : Carries blood towards the heart : : Pulmonary artery : _____.

(1) Carries blood towards right atrium

(2) Carries blood towards left atrium

(3) Carries blood away from the heart

(4) Carries blood towards right ventricle

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165. Match the items given in column I with those in column II and select the correct option given below.

	Column I		Column II
(a)	Tricuspid valve	(i)	Closes at the beginning of ventricular systole
(b)	Mitral valve	(ii)	Their closure produces second heart sound
(c)	Semilunar valves	(iii)	Between right atrium and right ventricle

(1) a(i), b(ii), c(iii)

(2) a(i), b(iii), c(ii)

(3) a(iii), b(i), c(ii)

(4) a(ii), b(i), c(iii)

166. Which of the following substances is not excreted through sweat?

(1) NaCl

(2) Urea

(3) Waxes

(4) Lactic acid

167. During hemodialysis, the principle used to remove urea from the patient's blood is

(1) Active transport of urea across a semi-permeable membrane into the dialyzing fluid

(2) Diffusion of urea down its concentration gradient from blood into dialyzing fluid

(3) Filtration of urea against its concentration gradients through a membrane similar to glomerulus filtration via active transport

(4) Osmosis of urea along the water potential gradient from dialysing fluid into blood

168. In humans, the counter current mechanism helps in concentrating the filtrate. This mechanism involves the direct participation of
(a) PCT
(b) Henle's loop
(c) DCT
(d) Vasa recta
Choose the **correct** option.

(1) (a) and (b)

(2) (c) and (d)

(3) (a) and (c)

(4) (b) and (d)

169. Human kidneys are reddish-brown structures situated between the levels of last _____ and third _____ vertebrae, close to the dorsal inner wall of the abdominal cavity.
Select the correct option to fill in the respective blanks.

(1) Lumbar; cervical

(2) Cervical; lumbar

(3) Lumbar; thoracic

(4) Thoracic; lumbar

170. The tubule/duct associated with kidney that allows passage of small amounts of urea into the medullary interstitium to maintain the osmolarity is

(1) PCT

(2) DCT

(3) Collecting duct

(4) Henle's Loop

171. Aldosterone aids in the maintenance of
a. Electrolytes
b. Body fluid volume
c. Osmotic pressure
d. Blood pressure
How many of the above factors hold true for the functions of aldosterone?

(1) Four

(2) Three

(3) Two

(4) One

172. Select a common character found in both red muscle fibres and white muscle fibres.

(1) Number of mitochondria

(2) Amount of sarcoplasmic reticulum

(3) Amount of myoglobin

(4) Presence of contractile proteins myosin and actin

173. Characteristic not applicable for skeletal muscle fibres is

(1) Structural syncytium

(2) Cylindrical appearance

(3) Presence of alternate dark and light bands

(4) Primarily associated with hollow visceral organs

174. Which of the following correctly depicts the order of events required for muscle contraction?
a. Impulse stimulates sarcoplasmic reticulum
b. Release of calcium ions
c. Muscle contraction
d. Nerve impulse reaches neuro-muscular junction
e. Cross-bridge formation

(1) a → d → b → e → c

(2) d → b → a → e → c

(3) d → a → b → c → e

(4) d → a → b → e → c

175. Osteoporosis, an age-related disease of skeletal system, may occur due to

(1) Immune disorder affecting neuromuscular junction leading to fatigue

(2) High concentration of Ca^{++} and Na^{+}

(3) Decreased levels of estrogen in menopausal females

(4) Accumulation of uric acid leading to inflammation of joints

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176. Which of the following pairs is **incorrectly** matched?

(1)	Ball and socket joint	-	Hip joint
(2)	Hinge joint	-	Between carpals and metacarpal of human thumb
(3)	Pivot joint	-	Between atlas and axis
(4)	Gliding joint	-	Between the carpals

(1) (1)

(2) (2)

(3) (3)

(4) (4)

177. Sella turcica, a depression enclosing the pituitary gland is found in

(1) Temporal bone

(2) Parietal bone

(3) Sphenoid bone

(4) Frontal bone

178. Match the column I with column II w.r.t total number of bones in adult humans and choose the **correct** option.

	Bone		**Number of bone**
a.	Limb bones	(i)	24
b.	Ribs	(ii)	14
c.	Facial bones	(iii)	06
d.	Girdle bones	(iv)	120

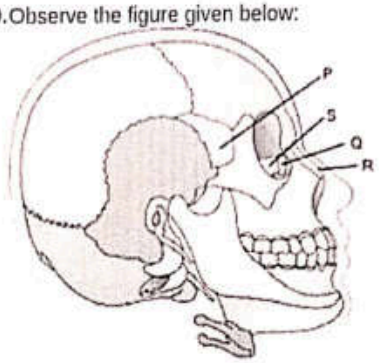
(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(i), c(ii), d(iii)

179. Observe the figure given below:



Select the **correct** option w.r.t labellings representing the different components of an adult human skull.

(1) P is a paired cranial bone that houses pituitary gland

(2) Q is an unpaired facial bone that is part of the eye socket.

(3) R is an unpaired facial bone just like Q that forms the bridge of the nose

(4) S is an unpaired cranial bone that forms the part of the roof of nasal cavity.

180. Rapid spasms in muscle due to low Ca^{++} in body fluid leads to

(1) Myasthenia gravis

(2) Muscular dystrophy

(3) Tetany

(4) Gout

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